Model Diagnostics

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Distance from trail + management

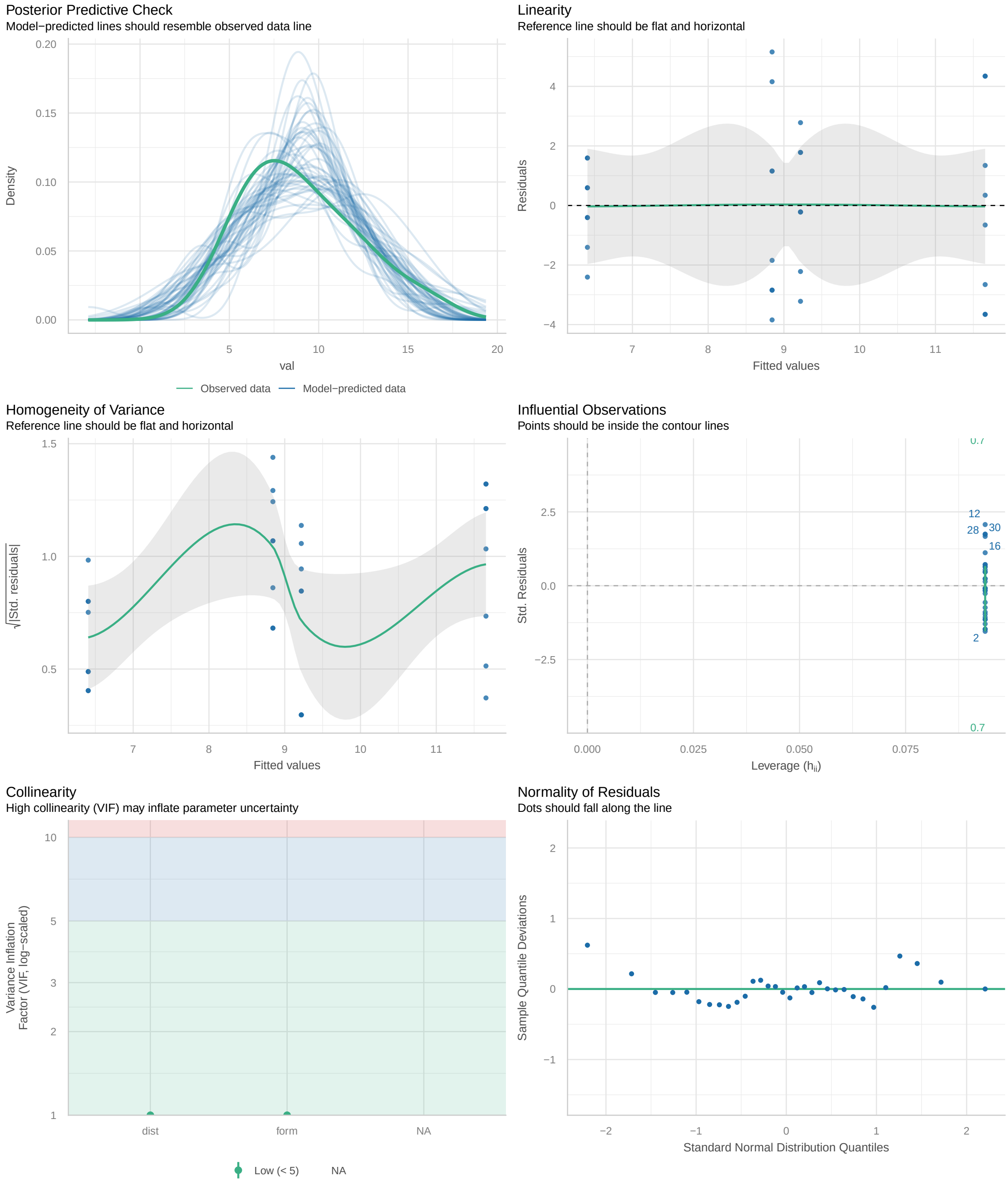
The models contained both distance from trail (close v far) and park management (formal vs informal):

$$y_i \sim \beta_0 + far_i * \beta_{far} + informal_i * \beta_{informal} + \epsilon_i$$

y_i was modelled as both species richness and Shannon.

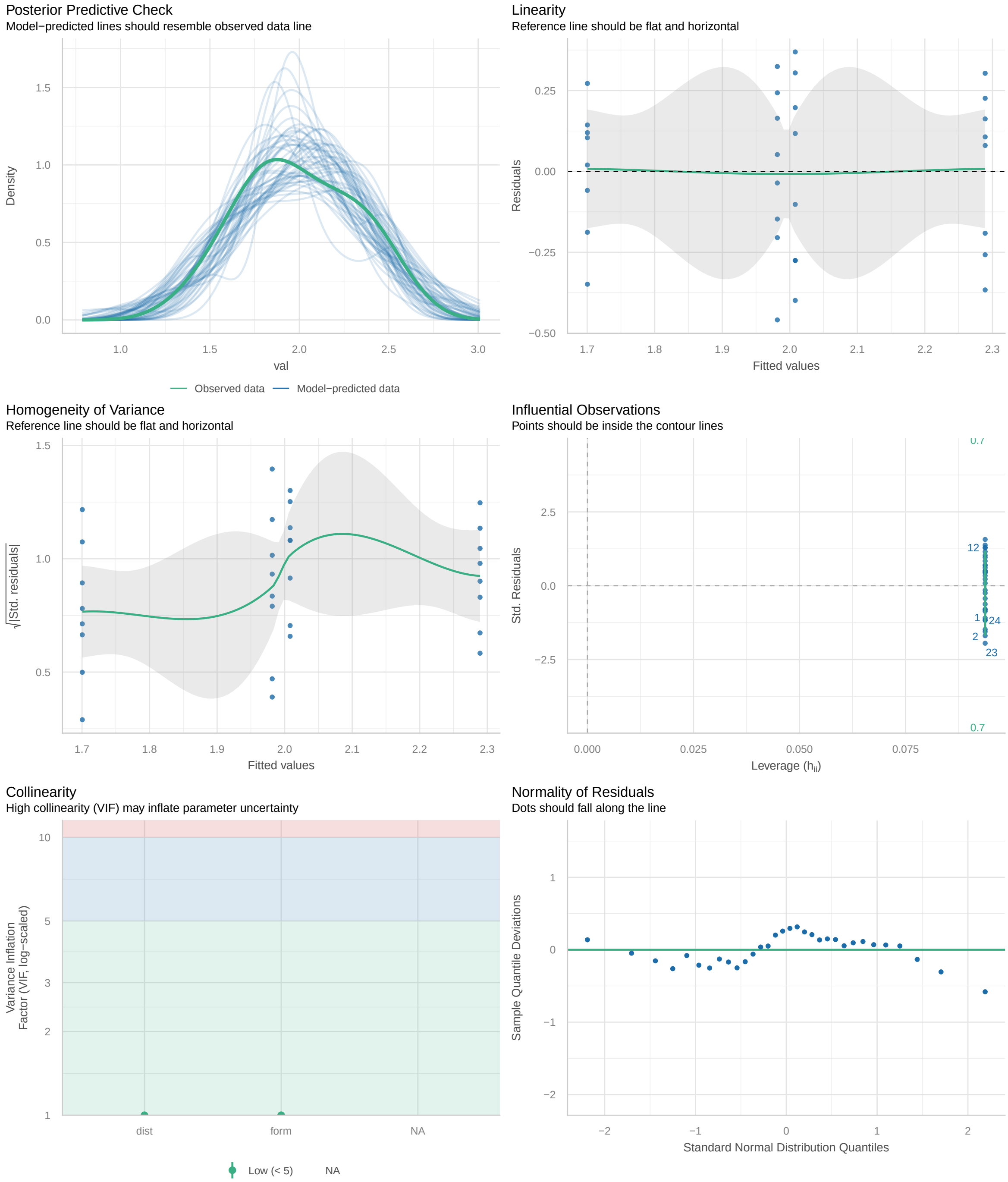
Species Richness

```
tar_load(SR_dist_model)

check_model(SR_dist_model)
```


Posterior Predictive Check
Model−predicted lines should resemble observed data line
Density
0.20
0.15
0.10
0.05
0.00
0
5
10
15
20
val
Observed data
Model−predicted data
Linearity
Reference line should be flat and horizontal
Residuals
4
2
0
−2
−4
7
8
9
10
11
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
√|Std. residuals|
1.5
1.0
0.5
7
8
9
10
11
Fitted values
Influential Observations
Points should be inside the contour lines
Std. Residuals
2.5
0.0
−2.5
12
28
30
16
2
0.7
0.000
0.025
0.050
0.075
Leverage (h_{ii})
Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log−scaled)
10
5
3
2
1
dist
form
NA
Low (< 5)
NA
Normality of Residuals
Dots should fall along the line
Sample Quantile Deviations
2
1
0
−1
−2
−1
0
1
2
Standard Normal Distribution Quantiles

Shannon

```
tar_load(shan_dist_model)

check_model(shan_dist_model)
```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Observed data
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Reference line should be flat and horizontal
Residuals
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
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Normality of Residuals
Dots should fall along the line
Sample Quantile Deviations
Standard Normal Distribution Quantiles

Vegetation

Species Richness

Shannon

Trail use

Species Richness

Shannon